

❖ **Nature and extent of damage:**

- The attack is more prevalent during monsoon.
- The wax moth larvae tunnel through the mid ribs of the comb and there is presence of small mass of minute wax particles outside the tunnels.
- In case of severe infestation, further brood rearing is stopped; bees stop field work and colony may abscond.

❖ **Prevention and control:**

- Close cracks and crevices in the hive. Reduce hive entrance.
- Remove combs not covered by bees. Keep the bottom board clean.

❖ **Control in storage**

- Keep spare combs in empty hive bodies in tiers and close both at bottom and top
- Disinfect the stack by burning sulphur @ 180 g/ cubic metre (fumigation by sulphur fumes).
- After fumigation, put naphthalene flakes in moth proof stacks

❖ **Ectoparasitic mites**

- In India, ectoparasitic mites *Varroa destructor* and *Tropilaelaps clareae* are causing severe damage to *A. mellifera* colonies.
- No damage in *A. cerana* colonies due to these mites.

Diseases in honeybees caused by

❖ **Spread of diseases**

- Most of the diseases spread among bees and from colony to colony or apiary to apiary through: Robbing, drifting, absconding, manipulation and migration; hence need proper management

❖ **Bacterial diseases**

❖ **European foul brood**

- First isolated report in 1971 in *Apis cerana*
- Late 1990's: in *A. mellifera* from North India; now widespread
- **Causative organism:** *Melissococcus pluton*
- Infected larvae in coiled stage; displaced in cells; die when 4-5 day old; dead brood mostly in uncapped stage.
- Colour of brood changes from shiny white to pale yellow then brown; on drying form rubbery scales.
- Sour Odour from dead brood; no ropiness in decaying larvae
- Worker, drone and queen; all type of brood affected
- **Control:** Feed Terramycin @200mg in 500 ml conc. sugar syrup to affected colony

❖ American foul brood

- **Causative organism:** *Paenibacillus larvae*
- **Time of death:** Late larval or early pupal stage
- **Position of dead brood:** Lying flat on cell base
- **Type of brood:** Worker, rarely drone or queen
- **Control:** Terramycin 0.25 – 0.4g in 5lt sugar syrup feeding

❖ Protozoa and mite diseases of adult bees

❖ Nosema; Protozoan disease

- **Nosema:** Severe during spring and winter
- Bees become dysentric, droppings of yellow excreta on combs or on parts of hive
- Infects all castes of bees; queen stops egg laying
- *Nosema apis* (protozoan)
- infected bees collect in front of hive, sluggish, crawlers on leaf blades, distended abdomen
- Feed fumigillin 200 mg in sugar syrup to each colony or two feedings at weekly interval of Dependel-M @ 0.5g/ litre/ colony